

1. In a function, users will enter the names of all of the students enrolled in a course, continuing to read them until the user enters an empty one. You'd like to return all of the names from the function.

List. The number of names is unknown and you need to append as the user enters them; lists are mutable, ordered, and allow duplicates.

2. Given an integer n , generate the first n integers in the Fibonacci sequence. The first integer in the sequence is 0, the second is 1, and each subsequent integer in the sequence is the sum of the two integers that precede it.

List. You need to build the sequence term by term as it is computed; a range cannot represent Fibonacci numbers, and a list is the natural mutable container for generated terms. A tuple could work after generation, but a list is simpler while generating.

3. Given two Cartesian points in either two-dimensional or three-dimensional space, determine the distance between them. Any two-dimensional point will be treated as though its z -coordinate is 0. What you'll need to decide is how to represent each point.

Tuple. A point is a small, fixed collection of coordinates, such as (x, y) or (x, y, z) . Tuples are immutable, ordered, and convenient for unpacking/indexing; a 2D point can be treated as $(x, y, 0)$.